AP BIOLOGY TOPIC 7.2

Natural Selection & Modes of Selection

UNIT 7: NATURAL SELECTION (13–20% EXAM WEIGHT)

EVO-1.E | **MODES OF SELECTION** | **PHENOTYPIC VARIATION**

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§1. KEY TERMS & FORMAL DEFINITIONS LEXICON — TOPIC 7.2 (NATURAL SELECTION & MODES OF SELECTION)

CED Core Vocabulary

Natural selection acts directly on phenotypes — (observable physical, biochemical, or behavioral traits), which indirectly changes underlying genotype and allele frequencies in the gene pool.

Sources of Variation — Genetic variation (random mutations, meiotic recombination/crossing over, independent assortment) + environmental influences.

Organisms don't "try" to adapt! — A giraffe doesn't stretch its neck and pass that on. Individuals with longer necks already existed; they just survived and reproduced more.

Fitness Impact — Different molecular variants can improve metabolic efficiency across temperatures, enhance pathogen protection, or confer toxin resistance.

Phenotype = Genotype + Environment — Traits that interact with the environment determine survival and reproduction.

Phenotypic Plasticity — The ability of a single genotype to produce different phenotypes under different environmental conditions. Helps short-term survival , but because it is not a...

Molecular Variation — Variation occurs at the cellular and molecular level in enzymes, pigments, membrane receptors, and surface proteins .

Genetic Variation Increases Adaptability — Populations with greater molecular and genetic variation have a wider range of possible responses to new selection pressures and higher chances of long-term...

§2. SELECTION ACTS ON PHENOTYPES (NOT DIRECTLY ON DNA)

Genotype × Environment

HIERARCHICAL CONCEPT & PATHWAY MAP **[CAUSE ➔ MECHANISM ➔ OUTCOME]**

Selection Acts on Phenotypes (Not Directly on DNA) ➔

Phenotype = Genotype + Environment ➔ **Sources of Variation** ➔ **Phenotypic Plasticity**

- **Natural selection acts directly on phenotypes** (observable physical, biochemical, or behavioral traits), which indirectly changes underlying **genotype and allele frequencies** in the gene pool.
- **Phenotype = Genotype + Environment:** Traits that interact with the environment determine survival and reproduction.
- **Sources of Variation:** Genetic variation (*random mutations, meiotic recombination/crossing over, independent assortment*) + environmental influences.
- **Phenotypic Plasticity:** The ability of a single genotype to produce different phenotypes under different environmental conditions. Helps **short-term survival**, but because it is not a genetic change, plasticity **does not directly cause evolution**.

[EXAMPLE & BIOLOGICAL APPLICATION] **Phenotype–Environment Interaction:** Poor childhood nutrition leads to reduced adult height despite inheriting alleles for tall stature; Arctic foxes change coat color from brown (summer) to white (winter) via phenotypic plasticity.

[CORE AXIOM / REMEMBER] **No variation = nothing to "select" from!**

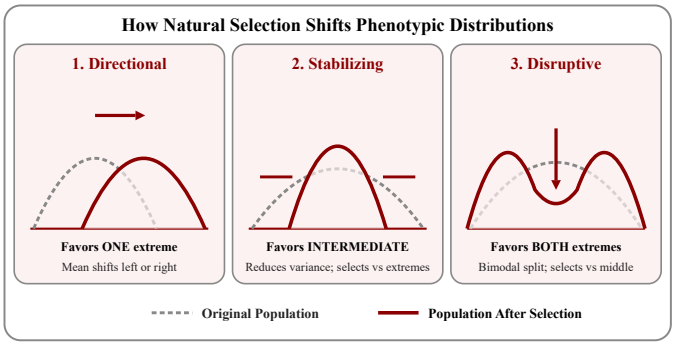
Without heritable phenotypic variation, natural selection cannot occur.

[COMMON AP EXAM FRQ TRAP] **Common Mistakes:**

- Assuming all individuals in a population respond the same to a selection pressure.
- **Organisms don't "try" to adapt!** A giraffe doesn't stretch its neck and pass that on. Individuals with longer necks already existed; they just survived and reproduced more.

§5. THREE MODES OF NATURAL SELECTION ON POLYGENIC TRAITS

High-Yield Graphs



SELECTION MODE	MECHANISM & DISTRIBUTION EFFECT	CLASSIC AP BIOLOGY EXAMPLES
Directional Selection	Favors phenotypes at **one extreme**; shifts population mean toward that extreme (common when environment changes).	• **Peppered moths (*Biston betularia*):** Shift from light to dark melanic form during Industrial Revolution soot pollution. • **Antibiotic/pesticide resistance** & Darwin's finch deeper beaks during drought.
Stabilizing Selection	Favors **intermediate phenotypes** and selects against both extremes; narrows distribution width and reduces genetic variance.	• **Human birth weight (~3.2–3.6 kg):** Very low-weight babies suffer higher mortality; very large babies cause birth complications. • **Bird clutch size:** Too few eggs = few offspring; too many = chicks starve.
Disruptive (Diversifying) Selection	Favors **both extremes** over intermediate phenotypes; creates a bimodal distribution and can drive sympatric divergence.	• **Black-bellied seedcrackers:** Small beaks crack soft seeds efficiently; large beaks crack hard seeds; intermediate beaks are inefficient at both.

[AP EXAM FRQ SCORING STRATEGY] **Graph Interpretation Tip:** On AP MCQs, compare the **dashed line (before)** to the **solid line (after)**: if the peak moves sideways → *Directional*; if the peak gets taller and narrower → *Stabilizing*; if the peak splits into two humps → *Disruptive*!

§3. MOLECULAR VARIATION WITHIN CELLS & ADAPTABILITY

Cellular Level

HIERARCHICAL CONCEPT & PATHWAY MAP **[CAUSE ➔ MECHANISM ➔ OUTCOME]**

Molecular Variation Within Cells & Adaptability ➔ **Molecular Variation** ➔ **Fitness Impact** ➔ **Genetic Variation Increases Adaptability**

- **Molecular Variation:** Variation occurs at the cellular and molecular level in **enzymes, pigments, membrane receptors, and surface proteins**.
- **Fitness Impact:** Different molecular variants can improve metabolic efficiency across temperatures, enhance pathogen protection, or confer toxin resistance.
- **Genetic Variation Increases Adaptability:** Populations with greater molecular and genetic variation have a wider range of possible responses to new selection pressures and higher chances of long-term survival.

[EXAMPLE & BIOLOGICAL APPLICATION] **Lactase Persistence:** Regulatory mutations that keep the *LCT* lactase enzyme gene expressed in adult humans provided a major nutritional and hydration advantage in pastoralist dairy-farming populations.

[CORE AXIOM / REMEMBER] **Genetic variation = biological toolbox for survival.**

§4. SEXUAL SELECTION & HETEROZYGOTE ADVANTAGE

Special Selection Modes

- **Sexual Selection:** Natural selection for traits that increase mating success, often producing pronounced **sexual dimorphism**:
  - **Intrasexual Selection:** Competition within the same sex (usually male-male combat, e.g., elk antlers, elephant seal territorial fights).
  - **Intersexual Selection (Mate Choice):** One sex (usually females) chooses mates based on showy ornaments or displays (e.g., peacock tail feathers, bird songs) that signal "good genes" despite predation trade-offs.
- **Heterozygote Advantage (Balancing Selection):** When the **heterozygous genotype (** **Aa** **)** has higher fitness than either homozygous genotype (**AA** or **aa**), maintaining both alleles in the population.

[EXAMPLE & BIOLOGICAL APPLICATION] **Sickle-Cell Trait (** **HbA/HbS** **) in Malaria Regions:**

- **HbA/HbA** (Normal): Susceptible to fatal *Plasmodium* malaria infection.
- **HbS/HbS** (Sickle-cell disease): Severe anemia / lower survival.
- **HbA/HbS** (Heterozygote): **Highest fitness** — resistant to malaria without severe sickle-cell disease!

§6. SCHOLARLY SYNTHESIS: EXPERIMENTAL DESIGN, QUANTITATIVE ANALYSIS, PATHOLOGY & AP FRQ RHETORIC

High-Yield Exam Rigor

MASTER TOPIC SYNTHESIS & CAUSAL ARCHITECTURE — TOPIC 7.2 **[MOLECULAR CAUSE ➔ CELLULAR MECHANISM ➔ PHYSIOLOGICAL / ECOLOGICAL EFFECT ➔ FRQ JUSTIFICATION]**

Topic 7.2: Natural Selection & Modes of Selection ➔ **Structural / Energetic Determinants** ➔ **Regulated Pathway & Homeostasis** ➔ **Experimental Perturbation (±2 SE_x̄)** ➔ **Phenotypic / Evolutionary Outcome**

SCHOLARLY DIMENSION	COMPREHENSIVE ACADEMIC REFERENCE & MECHANISTIC SPECIFICATION
I. Experimental Design & Required Lab Controls	AP Investigations 1, 2 & 3 (Artificial Selection, Hardy-Weinberg Mathematical Modeling & BLAST Bioinformatics): Comparing allele frequency shifts across generations under directional selection vs. small-population genetic drift simulations, and constructing maximum-parsimony cladograms from BLAST percentage sequence identity matrices (Cytochrome c / β-globin). **Null Hypothesis (H_0):** State explicitly that the independent variable treatment has *no statistically significant effect* on the dependent variable; any observed difference is due to random sampling chance alone.
II. Quantitative Formulas & Graph Interpretation	Quantitative Formulas & Golden Rule: Hardy-Weinberg Equilibrium (HWE): Allele frequencies $p + q = 1$; Genotype frequencies p^2 (homozygous dominant) + $2pq$ (heterozygous carriers) + q^2 (homozygous recessive) = 1. Golden Calculation Sequence: ALWAYS begin with the recessive phenotype frequency (q^2), take $\sqrt{(q^2)}$ to find q, compute $p = 1 - q$, and then calculate $2pq$ (heterozygote carrier frequency). Never take the square root of the dominant phenotype frequency ($p^2 + 2pq$)! **Error Bar Rule (±2 SE_x̄):** If the ±2 SE_x̄ error bars of two treatment groups *do not overlap*, there is a statistically significant difference between means ($p < 0.05$; reject H_0).
III. Pathology, Mutation & Disruption Case Study	Evolutionary & Biomedical Disruption Case Study: Incomplete antibiotic regimens or agricultural antibiotic overuse exert strong directional selection on bacterial populations (e.g., MRSA or MDR-Tuberculosis): susceptible cells die while rare, pre-existing resistant mutants (arising via random mutation or R-plasmid conjugation) survive and reproduce. In conservation biology, severe population bottlenecks (e.g., Northern Elephant Seals, African Cheetahs) drastically erode genetic diversity via genetic drift, increasing homozygosity for deleterious recessive alleles and vulnerability to novel pathogens.
IV. AP FRQ Task Verb Scoring Templates	• **DESCRIBE / IDENTIFY:** State the direct trend, value, or structural component with exact units from the figure. • **EXPLAIN (How/Why):** Connect cause to effect via a 3-link chain: *"[Change in X] causes [molecular/cellular mechanism Y], which results in [observable outcome Z]."* • **PREDICT:** State the specific directional change (increase, decrease, or remain constant) when the pathway is perturbed. • **JUSTIFY:** Pair quantitative data/evidence from the prompt with the underlying biological principle from Topic 7.2 to prove why the claim holds.

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